



Toegepaste Micro-economie: Public economics

Dr. Josse Delfgaauw

delfgaauw@ese.eur.nl

Week 7:
Collective decision-making

7. Collective Decision-making

So far, **normative** analysis of government intervention:

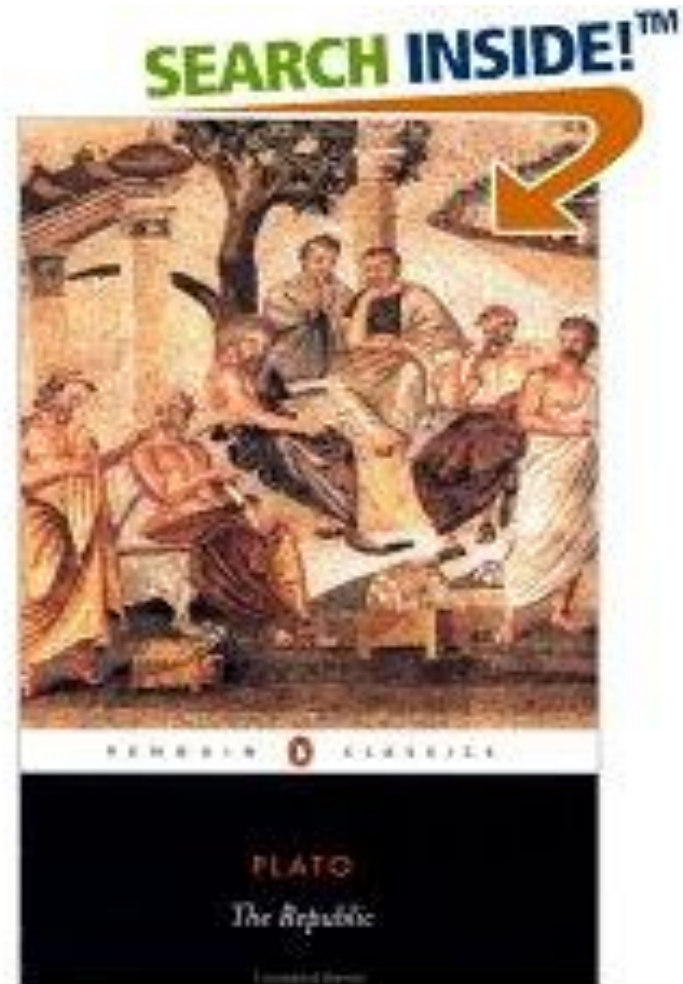
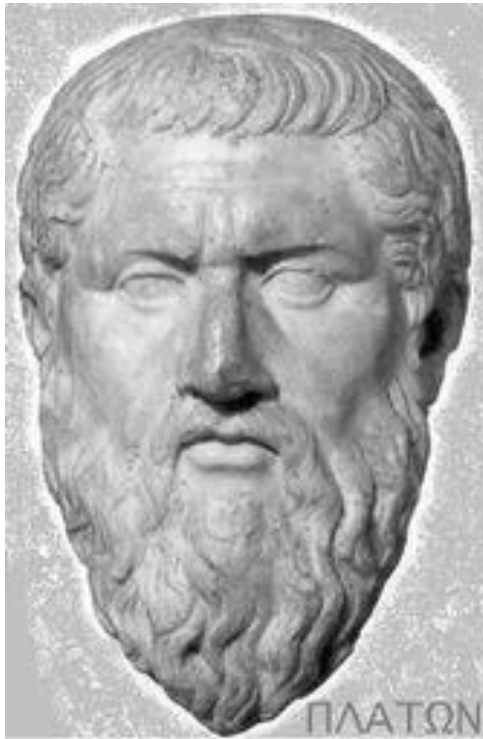
- What should be done, from a social welfare perspective?
- Takes government as ‘serving society’

Today: **positive** analysis of government intervention

- How are decisions made in practice?
- Starting point: decision-makers act in their own self-interest.
- Government failures

7. Collective Decision-making

The government as “benevolent dictator”, as in Plato’s *Politeia*?



7. Collective Decision-making

The government as perfect planner
(maximizing social welfare, no government failure)?



7. Collective Decision-making

The government's task is to do what the citizens ('the people') want it to do?



Collective Decision-making

Individual preferences for collective actions differ

- ‘The people’ does not exist.
- Almost all decisions have winners and losers
→ distribution!

Key question:

How to decide if different people want different things?

Additional problem:

People’s preferences (and firms’ production cost) are private

→ How to take all **true** preferences into account?

→ How to ‘aggregate’ preferences?

Government failure

Normative appeal of government intervention can be limited by problems in collective decision-making: government failures

Four main reasons for government failure:

1. Imperfect political representation and problems in aggregating preferences
2. Lack of information (about individuals' preferences and production processes)
3. Rent-seeking and corruption ('regulatory capture')
4. Limited / misaligned incentives in case of public production
→ Lack of financial (profit) motive makes public organisations less responsive to clients' needs (for better or worse!)

Roadmap

- Individual preferences for collective actions
- Collective decision-making rules / procedures
 - Direct democracy (voting over alternatives)
 - Arrow's Impossibility Theorem
 - Representative democracy
 - Politicians and elections
- Integrated Exercise
- Looking back on the course

Ind. preferences and collective actions

People differ in how much value they attach to goods and regulation provided by the government.

→ Public goods and publicly provided private goods, intervention in case of externalities and asym. info, redistribution, etc.

Result: people want different collective decisions / actions!

Note:

Preferences of citizens are generally not the same as social welfare function

→ Collective decision-making is often about making interpersonal trade-offs

Ind. preferences and collective actions

Example: suppose government spends G euro on public good, financed by (increase in) **proportional** tax t .

- Suppose: N individuals with Y_i and average income Y_m , then $G = tN Y_m$, such that $t = \frac{G}{NY_m}$

→ Individual taxes increase in income: $tY_i = \frac{GY_i}{NY_m}$

→ Marginal effect of G on taxes increases in own income: $\frac{\partial tY_i}{\partial G} = \frac{Y_i}{NY_m}$

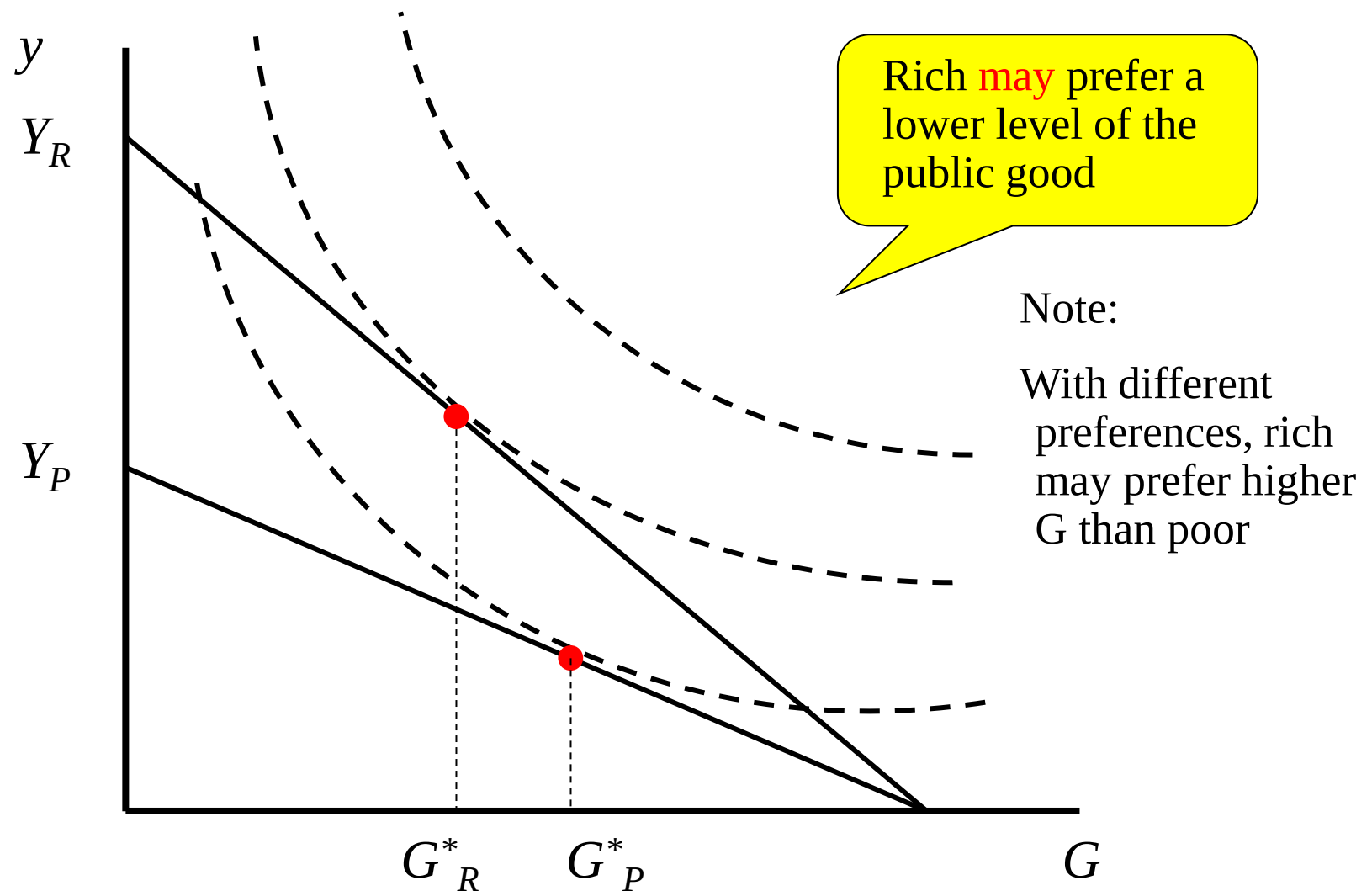
Consider twins: same preferences, different income

Higher income yields:

- Higher demand if public good is normal good
- Lower demand, due to higher ‘price’ per unit (= tax)

Result: rich people may prefer less G than poor.

Ind. preferences and collective actions



Decision-making procedures / rules

How to make decisions collectively?

- Given that people's preferences differ
→ How to 'aggregate preferences' ?

Some collective decision-making methods:

- Dictator ('social planner')
- Direct democracy ('referenda')
 - Unanimity
 - (Qualified) majority voting
- Representative democracy ('politicians')
- Constitutional democracy ('rule of law')

Decision-making procedures / rules

Direct democracy: everyone involved

- Unanimity rule:
 - Everyone should agree on decision. But how?
 - Recall: in case of public good, there is 1 efficient level of provision.

Theoretical solution: Lindahl prices

- Personalised tax shares

Problems:

- Strategic misrepresentation of preferences
- (Too) costly to determine individual tax shares

Decision-making procedures / rules

- (Qualified) majority voting: Deciding by majority
 - ‘Simple’ MV: 50% plus 1 vote
 - QMV: some other % (larger than 50%)
 - Used in several policy areas in the EU

Also problems:

- Repression of minorities
- Famous: Voting paradox (aka ‘Condorcet paradox’)

Example:

- Charles, Liz and Don collectively decide on where to go: soccer (S), bar (B), or movie (M)
- Preference orderings on next slide

Decision-making procedures / rules

	Charles	Liz	Don
First	S	B	M
Second	B	M	S
Third	M	S	B

Majority voting:

- Bar vs Soccer: 1-2
- Soccer vs Movie: 1-2
- Movie vs Bar: 1-2

→ Collective ordering:

- Bar < Soccer < Movie <
Bar < Soccer <

→ Ordering *not transitive!*

Result: cycling

Decision-making procedures / rules

Additional issues (see also exercise set 7):

1. Agenda setting power: Outcome depends on order of voting
2. Sincere voting versus strategic voting

In the example:

Ad 1. If Charles can set agenda, then ‘optimal’ order:

Round 1: Movie-Bar; Round 2: (Winner 1.)-Soccer

→ Outcome: soccer wins

Ad 2. Liz anticipates that soccer will win if she votes sincerely. Therefore, she strategically votes for Movie in first round. → Outcome: Movie wins

Note: voting paradox does not always arise

Decision-making procedures / rules

Obvious question: is there a ‘better’ procedure?

Related question: what are ‘reasonable’ criteria for a collective decision-making procedure?

- Unrestricted domain (all ind. preferences allowed)
- Pareto-criterium (If for all individuals $X \succcurlyeq Y$, then the collective ordering should have $X \succcurlyeq Y$)
- Non-dictatorship
- Transitivity (if $X \succcurlyeq Y$ and $Y \succcurlyeq Z$, then $X \succcurlyeq Z$)
- Independence of Irrelevant Alternatives
(if $X \succcurlyeq Y$ with choice set $\{X, Y\}$, then $X \succcurlyeq Y$ in choice set $\{X, Y, Z\}$)

Decision-making procedures / rules

Arrow's Impossibility Theorem:

It is not possible to find a decision-making procedure that *always* fulfils *all* these criteria.

Note: Theorem does *not* state that for a given decision, you cannot find such a procedure.

→ Simple example: with identical preferences, any rule works

Theorem implies that direct democracy cannot generally prevent 'unreasonable' outcomes

Decision-making procedures / rules

When does majority voting yield ‘stable’ outcome?

Answer: if individual preferences are ‘single-peaked’

- For each ind., one optimal outcome and options ‘further away’ from optimum are less and less desirable
- Note: violates first criterium

Outcome of MV: optimum of *median voter*

- Wins against all other options

Remarkable prediction:

- In a representative democracy with two parties who maximise prob. of winning elections, both parties ‘compete’ towards optimal policy for median voter

Decision-making procedures / rules

Conclusion direct democracy:

- Even if everyone is involved, outcome of any voting procedure may be unreasonable: Arrow's Impossibility T.
- Outcome may differ from socially optimal outcome

Two other downsides of direct democracy

- Costly
 - Particularly in large groups
- Limited incentives for individual voters to collect information on cost and benefits of options
 - Probability that an individual's vote is decisive is small
 - Particularly in large groups

Representative democracy

Decision-making contracted out to full-time decision-makers: politicians (and parties)

RD yields less costly decision-making, and leads (potentially) to better (more informed) decisions

Representative democracy

But: also creates problems:

- Politicians do not know preferences of all citizens
→ decisions may not reflect all preferences in society
- Politicians may not be willing to search for relevant information (or may misrepresent information for personal gain)
- Politicians (and government agencies) may make decisions that are in their self-interest (or the interest of only a subset of society, via corruption / lobbying)

Main institution to mitigate all these problems:

Elections (+ media)

Representative democracy

Focus on one problem: **Rent-seeking**

Some decisions involve billions of euro's

- Large infrastructural projects
- Mortgage interest deduction
- Climate change policies
- Gov support for firms / sectors during pandemic

When firms / small groups of citizens have large stakes, they have an incentive to spend resources on swaying the decision in their favour.

- Lobbying
- Corruption

Representative democracy

Why is rent-seeking a problem?

1. Resources used for unproductive activities
→ inefficient
2. May distort decision away from optimum
→ regulatory capture

A (potential) positive aspect: information collection

Summary

When people have different preferences for collective actions,
somehow decisions must be made

For small groups or important decisions

→ Direct democracy

But: Arrow's Impossibility Theorem

With large groups and many decisions:

→ Representative democracy

But: - politicians lack info on citizens' preferences

- politicians may be influenced / corruptible

→ Decisions may not be socially optimal

More on these issues: B3 course Political Economy

Summary

These issues contribute to government failure:

Groups / governments do not always manage to make decisions that maximise social welfare

→ Arrow's impossibility theorem

→ Lack of information

→ Limited / misaligned incentives of decision-makers

Consequence 1:

Trade-off between market failure and government failure

Consequence 2:

Good 'state capacity' crucial for proper functioning of markets

Integrated exercise

- Three equal-sized groups in society: rich (R), poor (P), middle-class (M); Income group i is Y_i
- Investment I (in million of euro's) in public transport network (public good)
- Paid for by tax (lump-sum, unless specified differently)
- Utility group i depends on consumption of 'composite good' y and I :

(note: $y_i = Y_i - \text{tax}$)

$$U_R = y_R + 4\sqrt{I}$$

$$U_M = y_M + 6\sqrt{I}$$

$$U_P = y_P + 10\sqrt{I}$$

Integrated exercise

First normative: what is socially efficient level of I ?

- Marginal cost = 1 (as utility decreases one-for-one with lump-sum tax for all groups)

- Marginal benefit:

$$MU_R = \frac{2}{\sqrt{I}} \quad MU_M = \frac{3}{\sqrt{I}} \quad MU_P = \frac{5}{\sqrt{I}}$$

- Social marginal benefit: $\sum_i MU_i = \frac{10}{\sqrt{I}}$

- $MSB = MSC$ gives $I = 100$

Integrated exercise

Now suppose: (not modeled) tax distortions, equally spread over the three groups

→ Marginal Cost of Public Funds (MCPF) = 1.2

MSB = MSC now gives

$$\frac{10}{\sqrt{I}} = 1.2$$

Which can be solved for $I = 69.44$

Integrated exercise

From now on, neglect tax distortions

Tax system is such that groups have different tax shares:

- Poor pay 1/6, Middle pay 1/3, Rich pay 1/2 of I
- Does this matter for efficient level of I ?
- No, given that there are no income effects (marginal utility of income = 1 for all groups)

But: groups do want different levels of I :

- Optimum for Rich: $\text{Max } U_R = Y_R - \frac{1}{2}I + 4\sqrt{I}$
- Which gives first-order condition $-\frac{1}{2} + \frac{2}{\sqrt{I}} = 0$
- Hence, $I^{R*} = 16$.
- Similarly, $I^{M*} = 81$ and $I^{P*} = 900$

Integrated exercise

Suppose SW-function has larger weight for utility of P than for M and R:

→ SW-maximizing level of $I > 100$.

Example: suppose

$$SW = \alpha_R U_R + \alpha_M U_M + \alpha_P U_P$$

$$\alpha_R = 1, \alpha_M = 1, \alpha_P = 3$$

Then $I = 225$ maximizes social welfare

Check yourself by maximizing

$$\max_I SW = Y_R - \frac{1}{2}I + 4\sqrt{I} + Y_M - \frac{1}{3}I + 6\sqrt{I} + 3\left(Y_R - \frac{1}{6}I + 10\sqrt{I}\right)$$

Integrated exercise

Now consider voting

Outcome of majority vote: $I = 81$

→ Preferences are single peaked (check!)

→ And middle-class is median voter

So, $I = 81$ wins against any alternative I

Hence, also outcome two-party representative democracy

→ not efficient and not SW-maximising!

Integrated exercise

Rent-seeking:

Rich want lower I

For reduction in I from 81 to 16, rich are willing to pay up to:

$$\begin{aligned} U_R(I = 16) - U_R(I = 81) &= (Y_R - \frac{1}{2}16 + 4\sqrt{16}) - (Y_R - \frac{1}{2}81 + 4\sqrt{81}) \\ &= 12.5 \end{aligned}$$

So, rich (maximally) willing to spend 12.5 million on ‘convincing’ politician to choose $I = 16$ rather than $I = 81$

Note: Rich willing to spend even more if politician would otherwise choose socially efficient $I = 100$ ($\rightarrow$ 18 million) or SW-maximizing $I = 225$ ($\rightarrow$ 60.5 million)

Looking back on today

Distinction between normative and positive analysis

Latter looks at decision-making process

- Individuals try to sway decisions in their favour
- Outcomes may depend on decision rule
- Helps to understanding why some ‘bad’ policies are undertaken (→ mortgage interest deduction) while some ‘good’ policies are not (→ flexible road pricing; emission tax; sugar tax)

Not just relevant for public decision-making; many issues also arise in (large) organisations

- Many decisions made by committees → collective decision-making!
- Non-productive efforts to increase probability of promotion is essentially rent-seeking
- More on these issues: major & master EMO

Looking back on the course

The normative (‘what should be done’) part looks at whether and how government intervention may improve upon market outcomes (from a social-welfare perspective).

→ efficiency and/or distribution

This course provides you with a unified framework for developing and evaluating policy proposals

Important elements:

- Market failure and government failure
- Cost-benefit analysis (including Excess Burden of taxes!)
- Efficiency – distribution trade-off

Beyond the course

Many issues discussed in public economics come back in major / master Policy Economics

- Labour economics
- Environmental economics
- Health economics (note: also separate major & master)
- (Advanced) economics of taxation
- Political economy
- Seminar economics of the public sector

Many insights relevant for (and used in) other fields as well:

→ Organisational economics (EMO); International Economics; (Corporate) finance / banking; Behavioural economics; Urban, regional & transport economics

Question hour

Lecture Friday: We answer your questions

E-mail your questions on/before Thursday:

- On personnel economics: dur@ese.eur.nl
- On public finance: delfgaauw@ese.eur.nl

Course evaluation:

Please take a moment to evaluate the course

We highly appreciate your feedback!

Good luck on the exam!!